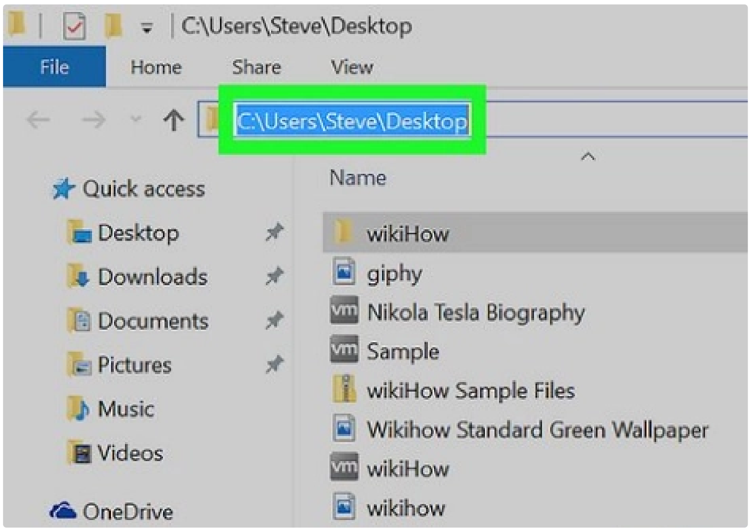
26F Stat TA Session W2

0. Preparation for TA session

- Please download the W2 example code and the attached datasets click here
    - the datasets can also be found in OpenIntro

1. Path/ Directory

- Path: the place where a file is saved



- Working directory: the place where `R` reads in your files
- Use `getwd()` to find your current working directory
- Use `setwd()` to change your working directory

⚠️ Note when you change the directory, remember to replace `\` with `/`

```
setwd("C:\your\working\directory") #Error! setwd("C:\\your\\working\\directory") #Works on Windows! setwd("C:/your/working/directory") #Works everywhere!
```

▼ Q. Why we can't simply copy the path and use `\` ?

Ans. Because `\` is an escape character in `R`

An escape character is used to represent special characters within a string

e.g.,

- \n: newline
- \t: tab

If you want to type `\` in `R` , use `\\`

2. Read Files

- The first step to perform statistical analysis is to read in files!
- There are several functions in `R` for reading different data formats

```
df <- read.csv("file.csv") #Work if file in working directory df <-
read.csv("folder_name/file.csv") #If file is in folder df <-
read.csv("C:/your/working/directory/file.csv") # example county <-
read.csv("county.csv") loan <- read.csv("loan50.csv")
```

3. What's in the dataset?

What the variables represent?

- `county.csv`
  - `county` : county name in the US
  - `homeownership` : home ownership rate during 2006-2010
  - `multi_unit` : percent of housing units in multi-unit structures, 2006-2010
- `loan50.csv`
  - `annual_income` : annual income of the applicant
  - `interest_rate` : interest rate of the loan the applicant received
  - `grade` : grade associated with the loan
- You can also find detailed descriptions in OpenIntro

What's hidden behind the numbers?

- You may want to first look at the numbers
    - What the average?
    - How dispersed the data is?
    - Is there any outliers? (離群值)
    - Is there any recording error in the dataset?
    - **Which method should we apply for further analysis?**
- One good way to get familiar with your dataset is to **draw figures** for variables of your interests
- To do this, we can use the package `ggplot2`

Scatter Plot `geom_point()`

- Check if two variables are related

```
ggplot(county, aes(x = multi_unit, y = homeownership)) + geom_point(color = "steelblue",
size = 1, # the arguments are to beautify your plot shape = 5 )
```

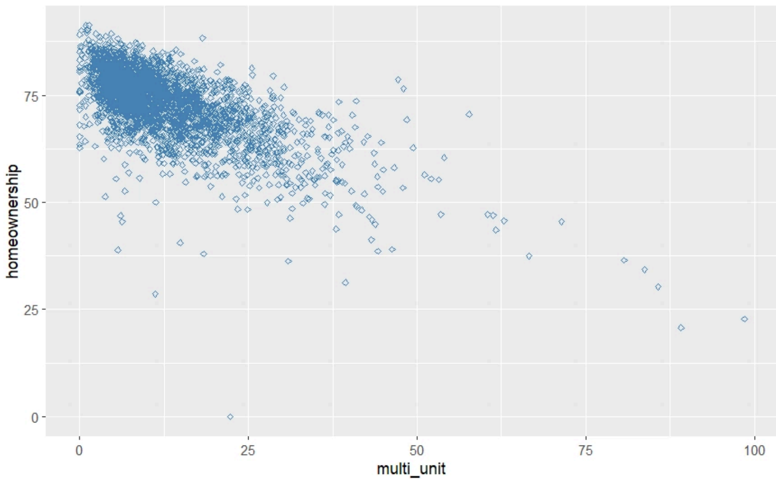


Figure: Scatter plot of multiple unit ratio and home ownerships

Histogram `geom_histogram()`

- How a variable is distributed? Is it skewed?

```
ggplot(loan, aes(x = interest_rate)) + geom_histogram(binwidth = 2.5, ## choose
appropriate bin width fill = "steelblue", color = "black" )
```

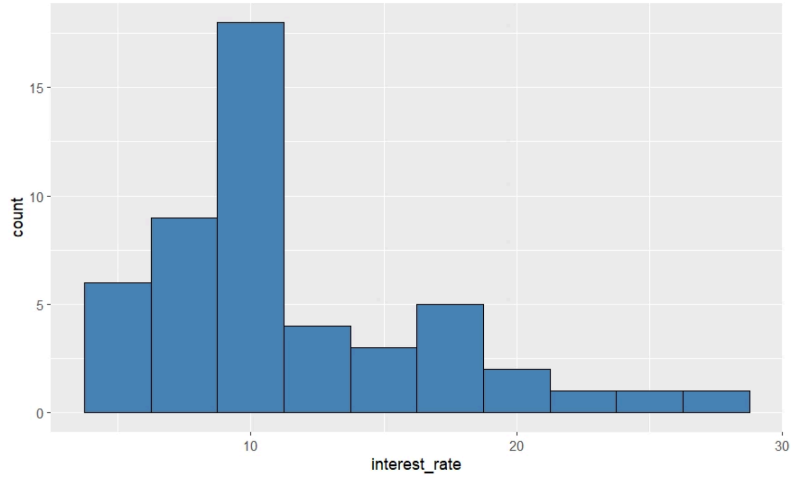


Figure: Histogram of interest rates

Box plot `geom_boxplot()`

- How dispersed a variable is? Is there any outliers?

```
ggplot(loan, aes(x = grade, y = interest_rate)) + geom_boxplot(width = 0.3)
```

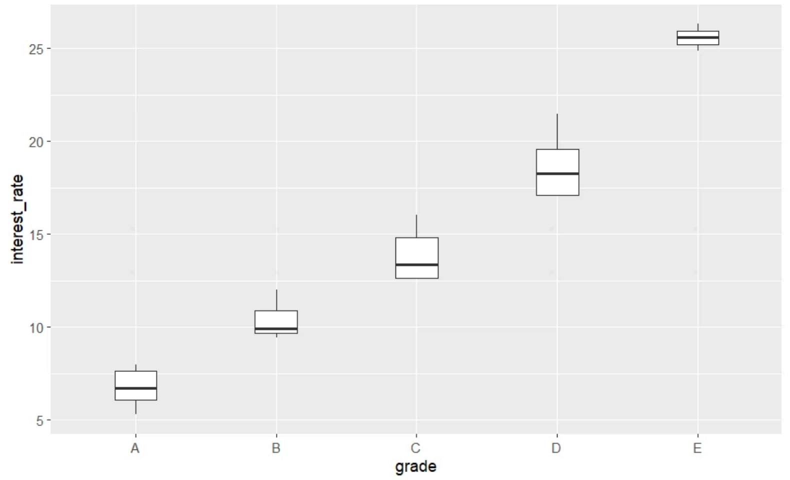


Figure: Box plot of interest rates by verified income

Dot plot `geom_dotplot()`

- Observe all data points, especially in small datasets

```
ggplot(loan, aes(x = grade, y = interest_rate)) + geom_dotplot(binaxis = 'y', stackdir =
'center', dotsize = 0.5, fill = "steelblue" )
```

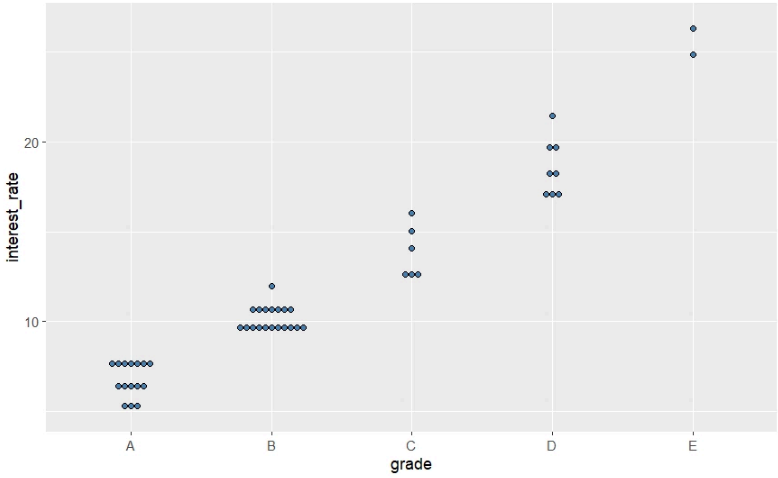


Figure: Dot plot of interest rates by verified income

Exercises

W2 Exercises